FH Salzburg

Wissenschaftliches Arbeiten 8

Wissenschaftliches Lesen

Technik
Gesundheit
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Wissenschaftliches Lesen

„Lesen ist Arbeit. Ein wissenschaftlicher Text – in seiner besonderen Form – erfordert auch eine besondere Art des Umgangs.“

(vor allem bei Texten, deren fachliche Begriffe und Diskurse uns bislang unbekannt waren!)

Vorrausetzung: Interesse & Zeit, eine leseleitende Perspektive & ein bestimmtes methodisches Vorgehen (Arbeitsschritte)



Software Engineering for Self-Adaptive Systems: A Second Research Roadmap (Draft Version of May 20, 2011)

Rogério de Lemos, Holger Giese, Hausi A. Müller, Mary Shaw
(Dagstuhl Seminar Organizer Authors)
r.delemos@kent.ac.uk, holger.giese@hpi.uni-potsdam.de, hausi@cs.uvic.ca, mary.shaw@cs.cmu.edu

Jesper Andersson, Luciano Baresi, Basil Becker, Nelly Bencomo, Yuriy Brun, Bojan Cukic, Ron Desmarais, Schahram Dustdar, Gregor Engels, Kurt Geihs, Karl M. Goeschka, Alessandra Gorla, Vincenzo Grassi, Paola Inverardi, Gabor Karsai, Jeff Kramer, Marin Litoiu, Antonia Lopes, Jeff Magee, Sam Malek, Serge Mankovskii, Raffaela Mirandola, John Mylopoulos, Oscar Nierstrasz, Mauro Pezzè, Christian Prehofer, Wilhelm Schäfer, Rick Schlichting, Bradley Schmerl, Dennis B. Smith, João P. Sousa, Gabriel Tamura, Ladan Tahvildari, Norha M. Villegas, Thomas Vogel, Danny Weyns, Kenny Wong, Jochen Wuttke
(Dagstuhl Seminar Participant Authors)

ABSTRACT

The goal of this roadmap paper is to summarize the state-of-the-art and identify research challenges when developing, deploying and managing self-adaptive software systems. Instead of dealing with a wide range of topics associated with the field, we focus on four essential topics of self-adaptation: design space for adaptive solutions, processes, from centralized to decentralized control, and practical run-time verification and validation. For each topic, we present an overview, suggest future directions, and focus on selected challenges. This paper complements and extends a previous roadmap on software engineering for self-adaptive systems published in 2009 covering a different set of topics, and reflecting in part on the previous paper. This roadmap is one of the many results of the Dagstuhl Seminar 10431 on *Software Engineering for Self-Adaptive Systems*, which took place in October 2010.

1. INTRODUCTION

The complexity of current software systems has led the software engineering community to investigate innovative ways of developing, deploying, managing and evolving software-intensive systems and services. In addition to the ever increasing complexity, software systems must become more versatile, flexible, resilient, dependable, energy-efficient, recoverable, customizable, configurable, and self-optimizing by adapting to changes that may occur in their operational contexts, environments and system requirements. Therefore, *self-adaptation* — systems that are able to modify their behavior and/or structure in response to their perception of the environment and the system itself, and their requirements — has become an important research topic in many diverse application areas.

It is important to emphasize that in all the many initiatives to explore self-adaption, the common element that enables its provision is usually software. Although software provides the required flexibility to attain self-adaptability, the proper realization of self-adaptation still remains a formidable intellectual challenge. Moreover, only recently have the first attempts been made to establish suitable software engineering approaches for the provision of self-adaptation. In the long run, we need to establish the foundations that enable the systematic development, deployment, management and evolution of future generations of self-adaptive software systems.

The goal of this roadmap paper is to summarize the state-of-the-art and identify research challenges when developing, deploying, managing and evolving self-adaptive software systems. Specifically, we focus on development methods, techniques, and tools that we believe are required when dealing with software-intensive systems that are self-adaptive in their nature. In contrast to merely speculative and conjectural visions and ad hoc approaches for systems supporting self-adaptability, the objective of this paper is to establish a roadmap for research and identify the key research challenges. Instead of dealing with a wide range of topics associated with the field, we focus on four essential topics of self-adaptation: design space of adaptive solutions, processes, from centralized to decentralized control, and practical run-

[1]This roadmap paper is a result of the Dagstuhl Seminar 10431 on *Software Engineering for Self-Adaptive Systems* in October 2010.

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Wann wurde es publiziert?

Wie verschaffe ich mir schnell einen Überblick?

Wie finde ich heraus ob das Paper qualitativ hochwertig ist?

Ziel: Ist der Artikel für MEINE wissenschaftliche Weiterarbeit brauchbar?

Blick auf:

* Abstract
* Einleitung
* Schluss / Conclusio
* Literaturverzeichnis

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